

Legend

checked named
result

checked displayed
equation

checked source
definition

added non-source
restriction

extension beyond
the printed claim

Status: partially formalized — all fifteen selected statements proved;
Proposition 3's strict claims need the added endpoint restriction $p^2 < 1$

Model, Section 4.1
 modes S, C, D with
 CE surpluses (1)–(3);
 menus $M_1 = \{S, C\} \subset$
 $M_2 = \{S, C, D\};$
 $Z^g = \mathbf{1}[\max_{m \in M_g} V^m \geq 0]$

Equation (4)
 $V^S \geq 0 \iff \omega \geq T^S$

Equation (5)
 $T^1 = T^S$
 $\max\{0, \gamma s - r^C\}$

Equation (6)
 $V^D \geq 0 \iff \omega \geq T^D$

Proposition 1
 $N_{it}^2 \geq N_{it}^1$ path by path

**Equation (18),
Assumption 1**
 $\gamma s - r^C \leq 0 \Rightarrow T^1 = T^S$

Equation (7)
 $B = \lambda[\alpha z - s\mu] - \kappa - r^D$
 $+ \frac{\rho}{2}[(2\lambda - \lambda^2)s^2/\pi - \sigma_D^2]$

Comparative static after (7)
 B monotone in a under
 Assumption 2
 (added $\rho, \lambda, z \geq 0$)

**Proposition 3,
Equation (10)**
 $\Delta C_i(s) \geq 0$ when $p^1 \leq p^2 \leq 1$

Proposition 2
 $Z^2 - Z^1 = \mathbf{1}[T^D \leq \omega < T^S]$
 under Assumption 1 and $B > 0$

Extension
 the band identity holds without $B > 0$
 (band empty, both sides zero)

Proposition 3, closed frontier
 strictly increasing
 and strictly concave
proved with added
 $0 < p^2 < 1, U_i \neq \emptyset;$
 at $p^2 = 1$: $\Delta C_i(s) = |U_i|$ for all s

**Proposition 4,
Equation (22)**
 $\sum_{k \in U_i} p = |U_i| p$; increasing in
 U_i and in ability

Proposition 5
 weak repository expansion;
 strict with one repository in the band